

Cisco ASA 5505 Firewall Security Implementation

Project Overview

This project demonstrates the implementation of a secure network using Cisco ASA 5505 Firewall in Cisco Packet Tracer.

The primary objective was to design, configure, and test a network environment where authorized services are allowed while unauthorized traffic is restricted through firewall security policies.

The project includes firewall configuration, Access Control Lists (ACLs), Network Address Translation (NAT), DNS services, web access verification, connectivity testing, and traffic filtering.

Objectives

The main objectives of this project were:

- Configure a Cisco ASA 5505 Firewall
- Create Inside and Outside security zones
- Implement Access Control Lists (ACLs)
- Configure Dynamic and Static NAT
- Allow HTTP and DNS traffic
- Block FTP traffic
- Verify network connectivity
- Test and validate firewall security policies
- Gain practical experience with network security technologies

Network Components

The network consists of the following devices:

Internal Network

- PC-1
- PC-2
- PC-3
- Internal Switch

Security Layer

- Cisco ASA 5505 Firewall

External Network

- Internet Router
- External Switch
- Web / DNS / FTP Server

The Cisco ASA Firewall acts as the security boundary between the internal and external networks.

IP Addressing Scheme

Device	IP Address
ASA Inside Interface	192.168.1.1/24
ASA Outside Interface	203.0.114.2/30
Internet Router	203.0.114.1/30
Server	203.0.113.10
PC-1	192.168.1.10
PC-2	192.168.1.11
PC-3	192.168.1.12

Firewall Configuration

The ASA firewall was configured with two security zones.

Inside Interface

- Name: INSIDE
- Security Level: 100
- IP Address: 192.168.1.1/24

Outside Interface

- Name: OUTSIDE
- Security Level: 0
- IP Address: 203.0.114.2/30

A default route was configured to allow communication with external networks.

Network Address Translation (NAT)

NAT was implemented to allow communication between private and public networks.

Dynamic NAT

Dynamic NAT was configured for the internal network:

- Internal Network: 192.168.1.0/24
- Translation Interface: OUTSIDE

This allows internal hosts to access external resources.

Static NAT

Static NAT was configured for:

- PC-1 (192.168.1.10)
- PC-2 (192.168.1.11)
- PC-3 (192.168.1.12)

Static mappings ensure consistent address translation and support communication initiated from outside the network.

Access Control Lists (ACLs)

ACLs were implemented to control network traffic and enforce security policies.

Permitted Traffic

Protocol	Purpose
HTTP	Web Access
DNS	Domain Resolution
ICMP	Connectivity Testing

Denied Traffic

Protocol	Purpose
FTP	Restricted by Firewall Policy

Example Policy:

- Allow HTTP traffic
- Allow DNS traffic
- Allow ICMP traffic

- Block FTP traffic
- Deny all other unauthorized traffic

Server Configuration

The server was configured with multiple services for testing purposes.

HTTP Service

A web service was hosted on:

netsecure.lk

DNS Service

Domain name resolution was configured for:

netsecure.lk - 203.0.113.10

FTP Service

FTP was enabled to verify that firewall policies correctly block unauthorized traffic.

Testing and Verification

Several tests were performed to verify network functionality and firewall behavior.

Connectivity Testing

Communication was verified between:

- PCs and Router
- Router and PCs
- Internal Network and Server

Result: Successful

HTTP Access Test

HTTP traffic was tested by accessing the hosted website.

Result: Successful

The firewall correctly permitted HTTP traffic according to the configured policy.

DNS Resolution Test

Domain name resolution was tested using the configured DNS service.

Result: Successful

The domain netsecure.lk was successfully resolved.

FTP Blocking Test

FTP connection attempts were made from internal hosts.

Result: Blocked

The firewall successfully denied FTP traffic according to the configured ACL rules.

NAT Verification

NAT functionality was verified using:

show nat

show xlate

Result: Successful

Address translations were created and verified correctly.

ACL Verification

Firewall rules were verified using:

show access-list

ACL hit counts confirmed that:

- HTTP traffic was permitted
- DNS traffic was permitted
- ICMP traffic was permitted
- FTP traffic was denied

Result: Successful

Challenges Encountered

Subnet Overlap Issue

Initially, overlapping subnets caused routing problems.

Solution:

The addressing scheme was redesigned and a dedicated subnet was used between the ASA firewall and the router.

NAT Configuration Issue

Older NAT commands were not supported by the Packet Tracer ASA version.

Solution:

Object-based NAT configuration was implemented.

DNS and HTTP Connectivity Issues

DNS and web access initially failed because of ACL configuration issues.

Solution:

Firewall rules were reviewed and adjusted to allow required services while maintaining security.

Router-to-PC Communication Issue

Communication initiated from the router to internal PCs was unsuccessful when only Dynamic NAT was configured.

Solution:

Static NAT entries were added for the internal hosts.

Access Information

The following credentials are provided for educational and testing purposes within the Packet Tracer environment.

Device	Credential Type	Password
Cisco ASA Firewall	Enable Password	cisco@NS\$28
Internet ASA Router	Console Password	cisco@NS\$27
Internet Router	Enable Secret Password	cisco@NS\$26

Skills Demonstrated

This project demonstrates practical experience with:

- Cisco ASA Firewall Configuration
- Network Security
- Access Control Lists (ACLs)
- Dynamic NAT
- Static NAT
- Routing Fundamentals
- DNS Services
- HTTP Services
- Traffic Filtering
- Connectivity Testing
- Network Troubleshooting
- Security Policy Enforcement

Conclusion

This project successfully demonstrated the implementation of a secure network using Cisco ASA 5505 Firewall in Cisco Packet Tracer.

Firewall policies were configured to permit authorized services such as HTTP and DNS while restricting FTP traffic. NAT functionality, connectivity, and security controls were tested and verified successfully.

The project provided valuable hands-on experience in firewall administration, traffic filtering, network troubleshooting, and practical network security implementation.